



Quantum-secured cash withdrawal using a QTM card

Quantum Banking System

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Description. This project-based course helps learners understand quantum computing through real-world everyday applications. The course focuses on how traditional banking systems can be transformed into quantum-secured systems using concepts from quantum algorithm. It also explores how security protocols can be designed and implemented using quantum gates and qubits within a quantum computing environment.

Today's banking systems mainly rely on classical security methods for protecting transactions and user data. However, with the rapid advancement of quantum computing capabilities, many existing security techniques may become vulnerable in the future. This creates the need for developing new quantum-secured system that can strengthen the security of classical banking systems.

Through this course, learners will study quantum security concepts, quantum gates, quantum circuits, and hybrid classical-quantum banking systems using simulations and practical examples. The course also demonstrates how such security problems can be modeled, programmed, and validated using quantum gates and circuits.

In addition, the course includes lecture slides, simulations, and GitHub resources to support hands-on learning and practical implementation.

Keywords. Banking, Cryptography, Quantum computing.

PACS. 03.67.-a, 03.67.Ac, 03.67.Dd, 89.20.Ff, 89.65.Gh.

Tools and Technologies. Python, Qiskit, Quantum circuits, Quantum gates, GitHub.

Intended Audience. Students, Researchers, Learners, Quantum computing enthusiasts.

Resource Type. Educational resource / Research project.

Version. 1.0

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GitHub. <https://github.com/QuantuMachine/Quantum-Secured-Cash-Withdrawal-Using-a-QTM-Card->

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